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Lab Report on:

# Intake Design

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# 1 Introduction

Supersonic inlets are crucial for air-breathing propulsion systems such as ramjets and scramjets, as they slow down the incoming supersonic or hypersonic freestream to subsonic conditions at the engine face while minimizing total pressure losses. In this study, a **single-ramp external compression inlet** operating at Mach 4.3 is designed and analyzed. The inlet employs a ramp deflection angle of $\theta = 11°$, which generates an oblique shock, followed by a reflected shock at the cowl lip and a terminal normal shock near the throat to complete the compression process. This configuration balances geometric simplicity with effective pressure recovery, making it suitable for hypersonic intake applications.

The design input parameters include a freestream Mach number of $M_\infty = 4.3$, ramp deflection angle $\theta = 11°$, mass flow rate $\dot{m} = 25\,\text{kg/s}$, freestream static temperature $T_\infty = 230\,\text{K}$, freestream static pressure $P_\infty = 1000\,\text{Pa}$, ratio of specific heats $\gamma = 1.4$, and gas constant $R = 287\,\text{J/(kg·K)}$. Based on these inputs, the performance evaluation shows a compression ratio of 2.8972, density ratio of 2.0662, and temperature ratio of 1.4022. The total pressure recovery is 0.8874, corresponding to a total pressure loss of 11.26%.

These results demonstrate that the inlet achieves effective compression with moderate increases in density and temperature, while preserving nearly 89% of the upstream stagnation pressure. The analytical framework, supported by Python implementation and validated using Ansys CFD simulations, confirms the reliability of the design and highlights its suitability for hypersonic propulsion applications.

# 2 Theoretical Background

## 2.1 Oblique Shock Theory

When a supersonic flow is deflected by a solid surface through an angle $\theta$ (the wedge or ramp angle), an oblique shock forms at an angle $\beta$ to the incoming flow. The relationship between these two angles and the freestream Mach number $M_1$ is given by the **Theta-Beta-Mach (TBM) relation**:

$$\tan\theta = \frac{2\cot\beta\left(M_1^2\sin^2\beta - 1\right)}{M_1^2\left(\gamma + \cos 2\beta\right) + 2} \tag{1}$$

For a given $M_1$ and $\theta$, Eq. (1) is transcendental in $\beta$ and must be solved numerically. Two solutions exist—the *weak shock* (lower $\beta$) and the *strong shock*—and the weak solution is the physically relevant one for inlet design.

### *2.1.1 Normal Mach Number Components*

The oblique shock is fully characterised by the *normal* component of the Mach number:

$$M_{n1} = M_1\sin\beta \tag{2}$$

Downstream of the shock, the normal Mach number is:

$$M_{n2} = \sqrt{\frac{1 + \dfrac{\gamma-1}{2}M_{n1}^2}{\gamma M_{n1}^2 - \dfrac{\gamma-1}{2}}} \tag{3}$$

The *total* downstream Mach number is then recovered by accounting for the flow deflection:

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)} \tag{4}$$

### 2.2 Rankine-Hugoniot Jump Conditions

Across the oblique shock, the static property ratios are determined entirely by $M_{n1}$, in exact analogy with a normal shock of the same normal Mach number:

$$\frac{P_2}{P_1} = 1 + \frac{2\gamma}{\gamma+1}\left(M_{n1}^2 - 1\right) \tag{5}$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_{n1}^2}{2 + (\gamma-1)M_{n1}^2} \tag{6}$$

$$\frac{T_2}{T_1} = \frac{P_2/P_1}{\rho_2/\rho_1} \tag{7}$$

### 2.3 Total Pressure Recovery

The total pressure recovery (TPR) is the ratio of stagnation pressure downstream to upstream. It quantifies the aerodynamic quality of the inlet—values close to unity indicate low losses:

$$\text{TPR} = \frac{P_{0,2}}{P_{0,1}} = \left[\frac{(\gamma+1)M_{n1}^2}{(\gamma-1)M_{n1}^2 + 2}\right]^{\gamma/(\gamma-1)} \cdot \left[\frac{\gamma+1}{2\gamma M_{n1}^2 - (\gamma-1)}\right]^{1/(\gamma-1)} \tag{8}$$

A total pressure loss of $(1 - \text{TPR}) \times 100\%$ is directly attributable to the entropy rise across the shock.

### 2.4 Isentropic Area–Mach Relation

Inside the convergent-divergent duct, flow is assumed isentropic. The ratio of duct cross-sectional area $A$ to sonic area $A^*$ is:

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M^2\right)\right]^{(\gamma+1)/[2(\gamma-1)]} \tag{9}$$

Applying Eq. (9) at both $M_1$ and $M_2$ gives the throat (sonic) area $A^*$ common to both stations, and hence the throat height $A_2$ (per unit span):

$$A_2 = \frac{AR(M_2)}{AR(M_1)} \cdot A_1 \tag{10}$$

where $AR(M)$ denotes the right-hand side of Eq. (9).

### 2.5 Geometry Relations

The capture height $A_1$ follows from continuity:

$$A_1 = \frac{\dot{m}}{\rho_\infty V_\infty} \tag{11}$$

where $\rho_\infty = P_\infty/(RT_\infty)$ and $V_\infty = M_\infty\sqrt{\gamma RT_\infty}$.

The ramp horizontal length $L$ is set by the vertical drop from $A_1$ to $A_2$:

$$L = \frac{A_1 - A_2}{\tan\theta} \tag{12}$$

The *cowl lip* must be positioned at the point where the oblique shock from the ramp leading edge intersects the cowl plane (shock-on-lip condition):

$$L_c = \frac{A_1}{\tan\beta} \tag{13}$$

## Intake Design Procedure

The intake was designed following a systematic step-by-step approach, outlined as follows:

1. **Selection of Ramp Angle**
   A ramp angle of $\theta = 11^\circ$ was selected to provide adequate flow compression while ensuring that the oblique shock remains attached.

2. **Normal Component of Upstream Mach Number**
   The normal component of the freestream Mach number ahead of the oblique shock was determined using:
   $$M_{n1} = M_1 \sin\beta$$

3. **Post-Oblique Shock Mach Numbers**
   Using oblique shock relations, the downstream normal and absolute Mach numbers were obtained as:
   $$M_{n2} = 0.6623, \quad M_2 = 3.4281$$

4. **Normal Shock at the Cowl Lip**
   A normal shock was assumed at the cowl lip to decelerate the flow to subsonic conditions. Standard normal shock relations were applied to determine the flow properties at the throat (Station 2).

5. **Thermodynamic Property Ratios**
   The resulting flow property ratios across the shock system are:

   - Static pressure ratio: 2.8972
   - Density ratio: 2.0662
   - Temperature ratio: 1.4022

6. **Total Pressure Recovery**
   The intake performance was evaluated in terms of total pressure recovery:
   $$\text{TPR} = 0.8874$$
   corresponding to a total pressure loss of 11.2559%.

7. **Geometric Parameters from Continuity**
   The inlet and throat areas were determined using the continuity equation:

$$\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

   which yielded:

$$A_1 = 1.2625\,\mathrm{m}^2, \quad A_2 = 0.5744\,\mathrm{m}^2$$

8. **Ramp Length Determination**
   Based on the selected geometry and the shock-on-lip condition, the ramp length was calculated as:

$$L_r = 3.5399\,\mathrm{m}$$

# 3 Input Parameters

Table 1: Design input parameters and their physical significance

| Symbol | Parameter | Value | Units |
|---|---|---|---|
| $M_\infty$ | Freestream Mach number | 4.3 | – |
| $\theta$ | Ramp deflection angle | 11 | ° |
| $\dot{m}$ | Design mass flow rate | 25 | kg/s |
| $T_\infty$ | Freestream static temperature | 230 | K |
| $P_\infty$ | Freestream static pressure | 1000 | Pa |
| $\gamma$ | Ratio of specific heats | 1.4 | – |
| $R$ | Specific gas constant (air) | 287 | J/(kg·K) |

The freestream conditions ($T_\infty = 230\,\mathrm{K}$, $P_\infty = 1000\,\mathrm{Pa}$) are representative of approximately 25 km altitude in the International Standard Atmosphere—a typical cruise altitude for a Mach 4.3 vehicle.

Table 2: Performance Parameters

| Parameter | Value |
|---|---|
| Mass flow rate, $\dot{m}$ | 25 kg/s |
| Compression ratio | 2.8972 |
| Density ratio | 2.0662 |
| Temperature ratio | 1.4022 |
| Total pressure recovery (TPR) | 0.8874 |
| Total pressure loss | 11.2559% |

The intake's performance outcomes. It shows that the system processes a mass flow rate of 25 kg/s, achieving a compression ratio of 2.8972, density ratio of 2.0662, and temperature ratio of 1.4022. With a total pressure recovery of 0.8874 and a loss of 11.26%, the intake demonstrates efficient compression while maintaining good pressure recovery for downstream operation.

# 4 Design Considerations and Limitations

## 4.1 Shock-on-Lip Condition

The cowl is placed at $x = L_c$ (Eq. 13) so that the oblique shock passes exactly through the cowl lip. This *shock-on-lip* design point maximises mass capture: at lower Mach numbers the shock would fall inside the cowl (spillage), while at higher Mach numbers it falls outside (over-capture with increased drag).

## 4.2 Two-Dimensional Assumption

The current model is two-dimensional (per unit span). For an axisymmetric inlet (cone) the governing relation is the *Taylor–Maccoll equation*, which produces a conical shock at a slightly different angle for the same turning. Area ratios must then use annular areas ($A = \pi r^2$).

## 4.3 Viscous and Real-Gas Effects

Boundary layer growth on the ramp surface displaces the effective flow and reduces capture mass flow. At Mach 4.3, the boundary layer momentum thickness can be significant (of order 1 % of $A_1$). Real-gas and high-temperature effects become important above Mach 7 and are not included here.

## 4.4 Multi-Ramp Extension

For better total pressure recovery, the inlet can be split into $n$ ramps of smaller turning angles. If the total turning is $\Theta$, then each ramp deflects by $\Theta/n$. Each successive shock is weaker, giving a higher cumulative TPR. The code is structured to extend naturally by iterating the shock calculation across multiple ramp stages.

# 5 Computational Implementation

## 5.1 Solving the Theta-Beta-Mach Relation

The TBM equation (Eq. 1) is rearranged to a root-finding form $f(\beta) = 0$:

$$f(\beta) = \frac{2\cot\beta\,(M_1^2\sin^2\beta - 1)}{M_1^2(\gamma + \cos 2\beta) + 2} - \tan\theta = 0 \tag{14}$$

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import fsolve
M1 = 4.3
theta_deg = 11
m_dot = 25
theta = np.deg2rad(theta_deg)
gamma = 1.4
T_inf = 230
P_inf = 1000
R = 287
def tbm(b):
    return (2 * (1/np.tan(b)) * ((M1**2 * (np.sin(b)**2)) - 1) /
            (M1**2 * (gamma + np.cos(2*b)) + 2)) - np.tan(theta)
```

```
15
16 mu = np.arcsin(1/M1)
17 beta_rad = fsolve(tbm, mu)[0]
18 beta = np.rad2deg(beta_rad)
```

Listing 1: TBM root-finding with `scipy.optimize.fsolve`

### 5.2 Oblique Shock Properties

```
1  Mn1 = M1 * np.sin(beta_rad)
2
3  Mn2 = np.sqrt((1 + (gamma-1)*0.5*Mn1**2) /
4                (gamma*Mn1**2 - (gamma-1)*0.5))
5
6  M2 = Mn2 / np.sin(beta_rad - theta)
7  Compression_ratio = 1 + (2*gamma)/(gamma+1)*(Mn1**2 - 1)  # P2/P1
8  Density_ratio = (gamma+1)*Mn1**2 / (2 + (gamma-1)*Mn1**2)  # rho2/rho1
9  Temperature_ratio = Compression_ratio / Density_ratio
10 TPR = (((gamma+1)*Mn1**2 / ((gamma-1)*Mn1**2 + 2))**(gamma/(gamma-1))) *
       \
11       (((gamma+1) / (2*gamma*Mn1**2 - (gamma-1)))**(1/(gamma-1)))
```

Listing 2: Post-shock Mach number and property ratios

### 5.3 Area Ratios and Throat Sizing

```
1  a = np.sqrt(gamma * R * T_inf)
2  v = a * M1
3  rho = P_inf / (R * T_inf)
4  L_f = 8
5  A_c = 0.5
6  A1 = m_dot / (rho * v)
7  AR1 = (1/M1) * (2/(gamma+1) * (1 + (gamma-1)*0.5*M1**2))**((gamma+1)
       /(2*(gamma-1)))
8  AR2 = (1/M2) * (2/(gamma+1) * (1 + (gamma-1)*0.5*M2**2))**((gamma+1)
       /(2*(gamma-1)))
9  A2 = (AR2 / AR1) * A1
10 L = (A1 - A2) / np.tan(theta)
11 L_r = L / np.cos(theta)
12 L_c = A1 / np.tan(beta_rad)
```

Listing 3: Isentropic area ratio function and throat height calculation

Python code: Google Drive Folder

# 6 Results

## 6.1 Intake Geometry

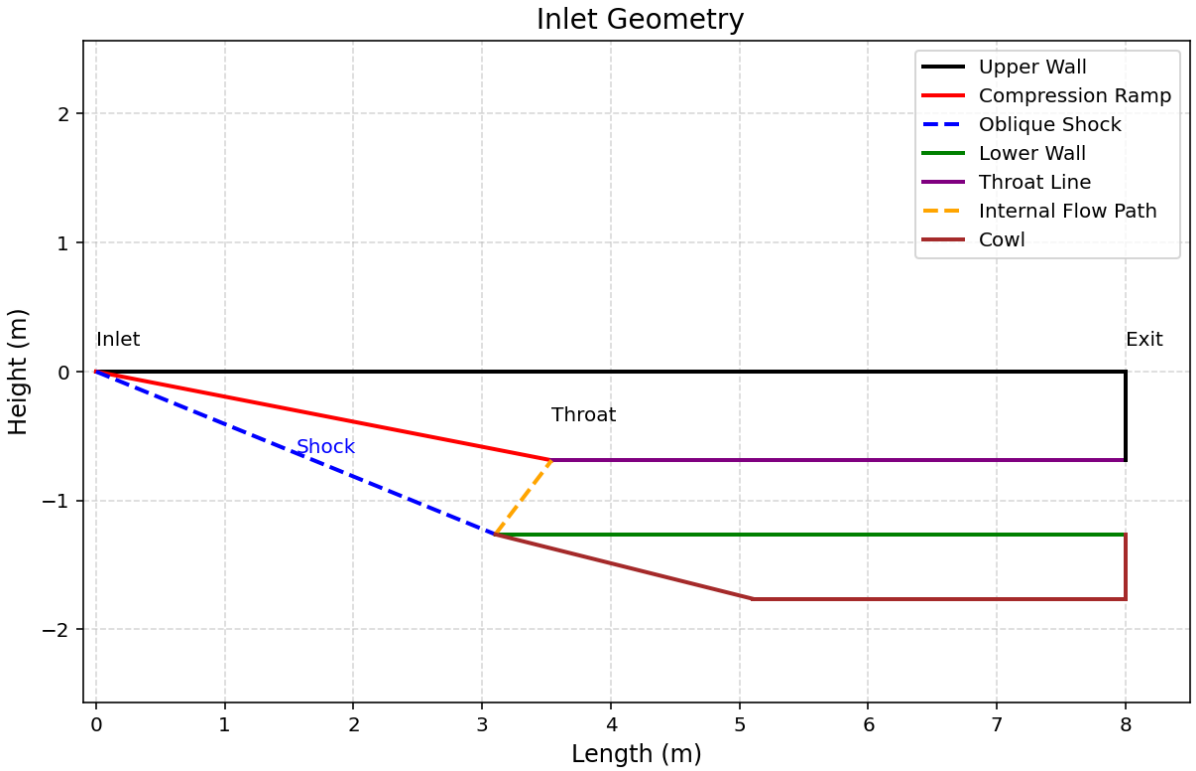



Figure 1: Intake geometry calculated and generated using Python:

# 7 Numerical Simulation

## 7.1 Validation of Analytical Design

To validate the analytical design and evaluate the real flow behavior, the designed hypersonic intake was tested using numerical simulations. The inlet configuration was based on the analytically determined parameters: freestream Mach number $M_\infty = 4.3$, ramp deflection angle $\theta = 11°$, design mass flow rate $\dot{m} = 25\,\text{kg/s}$, freestream static temperature $T_\infty = 230\,\text{K}$, freestream static pressure $P_\infty = 1000\,\text{Pa}$, ratio of specific heats $\gamma = 1.4$, and gas constant $R = 287\,\text{J/(kg·K)}$.

The resulting intake performance was characterized by a compression ratio of 2.8972, density ratio of 2.0662, and temperature ratio of 1.4022. The total pressure recovery was found to be 0.8874, corresponding to a total pressure loss of 11.26%. These simulation results were used to visualize the shock wave structure and flow field inside the intake, identify non-ideal effects such as shock–boundary layer interaction and shock detachment, and assess the overall performance under realistic viscous flow conditions.

## 7.2 Simulation Objectives

The numerical simulation was conducted to achieve the following objectives:

- To resolve and visualize the internal shock structure and overall flow field within the intake.
- To identify non-ideal flow phenomena, including shock–boundary layer interaction, flow non-uniformity, and potential shock detachment, which are not captured by inviscid analytical models.
- To evaluate the intake performance under realistic viscous flow conditions and assess deviations from theoretical predictions.

### 7.3 Simulation Setup

A two-dimensional representation of the hypersonic intake was constructed based on the analytically derived design parameters. The geometry incorporates a ramp angle of $\theta = 11^\circ$, a capture area of $A_1 = 1.2624\,\mathrm{m}^2$, a throat area of $A_2 = 0.5744\,\mathrm{m}^2$, and a ramp length of $L_r = 3.5399\,\mathrm{m}$.

A structured computational mesh was generated with targeted refinement in regions of strong gradients, particularly along the ramp surface, near the cowl lip, and within expected shock interaction zones. Inflation layers were applied along solid boundaries to adequately resolve the boundary layer and capture viscous effects. This meshing strategy ensures accurate prediction of shock structures and their interaction with the boundary layer, which is critical for hypersonic intake analysis.

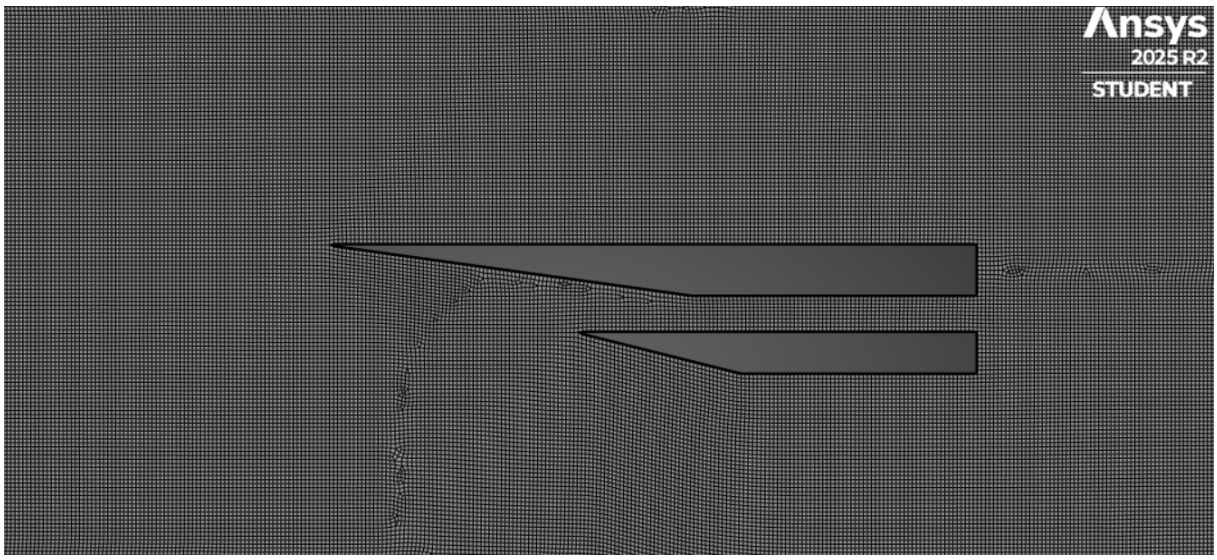



Figure 2: mesh

### 7.4 Governing Equations and Numerical Solver

The flow field was computed by solving the compressible Reynolds-Averaged Navier–Stokes (RANS) equations using a density-based formulation, which is appropriate for capturing strong compressibility effects in hypersonic regimes. Air was treated as a calorically perfect gas with a specific heat ratio of $\gamma = 1.4$.

To account for turbulence effects and their interaction with shock structures, the $k$–$\omega$ SST turbulence model was employed. This model provides reliable performance in regions with adverse pressure gradients and shock–boundary layer interactions, which are dominant features in hypersonic intake flows.

### 7.5 Boundary Condition Specification

The computational domain was defined using physically consistent boundary conditions representing flight and intake operating conditions:

- **Freestream inlet:** A pressure far-field boundary was imposed corresponding to a free-stream Mach number of $M_1 = 6$, with static pressure and temperature specified according to the target flight altitude conditions.
- **Outlet boundary:** A pressure outlet condition was prescribed, allowing the internal flow to adjust naturally while accommodating downstream pressure effects.
- **Solid surfaces:** All walls were modeled as no-slip and adiabatic, ensuring proper resolution of viscous effects without heat transfer.
- **Symmetry planes:** Where applicable, symmetry conditions were used to reduce computational cost without affecting flow fidelity.

### 7.6 Solution Convergence

Numerical convergence was established through both residual reduction and flow stability criteria. The solution was deemed converged when all governing residuals dropped below $10^{-6}$ and the mass flow rate between inlet and throat sections remained consistent within 0.5% variation over iterations.

## 8 CFD Results

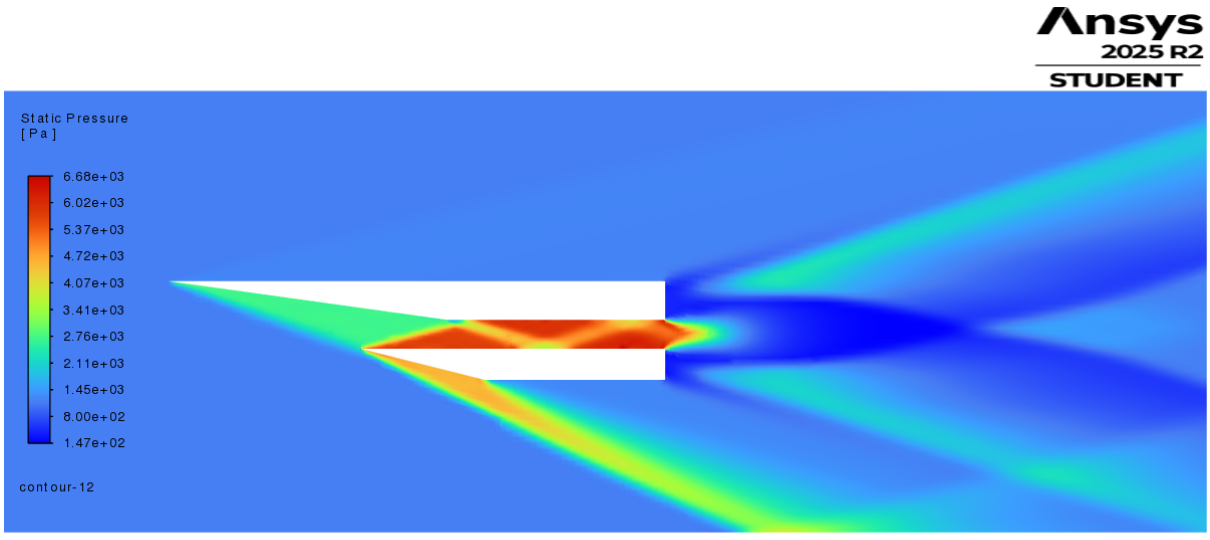



Figure 3: Static Pressure Contour

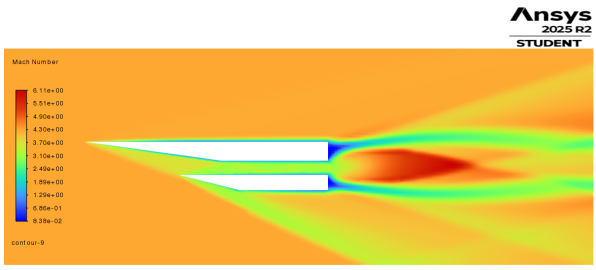



Figure 4: Mach Number Contour

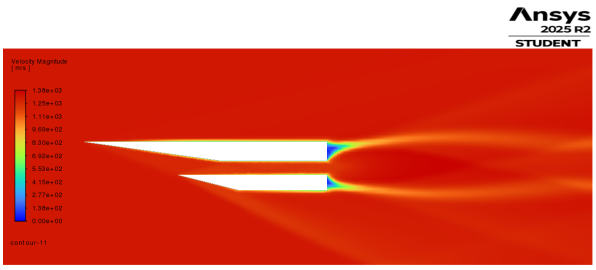



Figure 5: Velocity Magnitude Plot

## 9 CFD Results and Discussion

The intake configuration was examined through steady-state CFD simulations conducted in ANSYS Fluent. From the converged numerical solution, critical performance indicators—including the mass flow rate and total pressure recovery (TPR)—were extracted. These values were then compared directly with the analytical predictions to assess inlet efficiency under viscous flow conditions and to evaluate the fidelity of the theoretical design approach

The mass flow rate obtained from the CFD simulation is **25.8525 kg/s**, showing excellent agreement with the analytical design target of **25 kg/s**, with a deviation of **+1.0341%** (Table 3). This confirms that the selected capture area and ramp geometry effectively ingest the required airflow.

Table 3: Comparison of Mass Flow Rate between Analytical and CFD Results

| **Method** | **Mass Flow Rate (kg/s)** | **Deviation (%)** |
|---|---|---|
| Analytical (Oblique Shock Theory) | 25 | – |
| CFD Simulation (ANSYS Fluent) | 25.8525 | +1.0341 |

Total pressure recovery is a key performance metric for hypersonic intakes, as it directly influences overall propulsion efficiency. It is defined as:

$$TPR = \frac{P_{t,\,throat}}{P_{t,\,inlet}} \tag{15}$$

Using area-weighted average values obtained from ANSYS Fluent:

Table 4: Total Pressure Values and Recovery from CFD Simulation

| **Location** | **Area-Weighted Total Pressure (Pa)** |
|---|---|
| Freestream Inlet ($P_{t1}$) | 251834.35 |
| Throat / Engine Face ($P_{t2}$) | 213214.68 |
| Total Pressure Recovery (TPR) | 0.8466 |

The CFD-based total pressure recovery is:

$$TPR_{CFD} = \frac{2{,}51{,}834.35}{2{,}13{,}214.68} = 0.8466 \tag{16}$$

This value is lower than the analytical prediction of **0.8874**, obtained from inviscid oblique shock theory. The difference ($\Delta TPR \approx 0.0408$, or $\sim 4.6\%$ reduction) is expected and primarily attributed to:

- Viscous dissipation and boundary layer growth,
- Entropy generation across the lambda shock system,
- Local flow separation in high-pressure-gradient regions.

Table 5: Comparison between Analytical and CFD Results

| **Parameter** | **Analytical** | **CFD** |
|---|---|---|
| Mass Flow Rate (kg/s) | 25 | 25.8525 |
| Total Pressure Recovery (TPR) | 0.8874 | 0.8466 |
| Total Pressure Loss (%) | 11.256 | 15.34 |

The CFD simulations indicate excellent consistency with the analytical prediction in terms of mass flow rate, while the agreement in total pressure recovery is satisfactory. The observed reduction in TPR underscores the influence of viscous effects, Despite this discrepancy, the intake demonstrates reliable performance at the design Mach number, thereby confirming the validity of the analytical methodology and emphasizing the complementary role of CFD in capturing real-flow physics beyond idealized assumptions.

### 9.1 Conclusion

This study presented the analytical design and numerical validation of a two-dimensional supersonic inlet operating at a freestream Mach number of $M_\infty = 4.3$. The inlet was designed using classical compressible flow theory, including oblique shock relations, Rankine–Hugoniot jump conditions, and isentropic area–Mach relations. A single-ramp external compression configuration with a ramp angle of $\theta = 11^\circ$ was adopted to achieve the desired mass flow rate and pressure recovery characteristics while maintaining geometric simplicity. The analytical design predicted a mass flow rate of 25 kg/s with a total pressure recovery of 0.8874. These results were validated using steady-state CFD simulations in ANSYS Fluent. The numerical results showed a mass flow rate of 25.8525 kg/s, demonstrating excellent agreement with the analytical prediction, with a deviation of only +1.03%. The total pressure recovery obtained from CFD was 0.8466, which is lower than the analytical value by approximately 4.6%. This reduction is attributed to viscous effects, boundary layer growth, entropy generation, and shock–boundary layer interactions, none of which are captured in the inviscid analytical formulation.

The comparison between analytical and CFD results confirms that the theoretical design methodology provides a reliable first-order estimate of inlet performance. However, CFD analysis reveals important real-flow phenomena that significantly influence total pressure losses and overall efficiency. In particular, shock–boundary layer interaction and viscous dissipation play a dominant role in degrading total pressure recovery in hypersonic and high-supersonic regimes.

Overall, the designed inlet successfully meets its performance requirements under design conditions, demonstrating effective compression and stable mass capture. The study highlights the importance of combining analytical methods with high-fidelity numerical simulations to achieve accurate and physically realistic hypersonic intake designs. Future improvements may include multi-ramp configurations, three-dimensional effects, and optimization for enhanced total pressure recovery and reduced losses.

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## 10 Appendix:

Python code: Google Drive Folder